

Feiyang Zheng

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EDUCATION

North Carolina State University

Raleigh, NC

Ph.D. in Computer Science, GPA 3.75/4.3; advisors: Frank Mueller, Jiajia Li Aug 2025 – May 2030 (expected)

- Coursework: Architecture of Parallel Computers, Advanced Computer Architecture, Compiler Construction, Code Optimization; in progress: Numerical Analysis I, Software Engineering.

Southern University of Science and Technology

Shenzhen, China

B.S. in Mathematics and Applied Mathematics

Sep 2021 – Jun 2025

RESEARCH EXPERIENCE

Graduate Research Assistant

Aug 2025 – Present

North Carolina State University

Raleigh, NC

- Scientific data compression (NSF EPOD #2504512) and GPU program analysis; see *Selected Projects*.

Research Assistant

Sep 2024 – Jun 2025

Southern University of Science and Technology

Shenzhen, China

- Built a LibTorch-based C++ deep learning package for unknown-equation modeling: CMake build integrating CUDA and yaml-cpp, with C++ model templates designed for straightforward secondary development.

TEACHING EXPERIENCE

Teaching Assistant

Summer 2026 – Present

North Carolina State University

Raleigh, NC

- **CSC/ECE 506 Architecture of Parallel Computers** (Summer 2026): authored the programming assignments and part of the exam problems; graded and supported students.
- **CSC 514 Foundations of Cryptography** (Fall 2026).

PUBLICATIONS

Dogan Corus, Pietro S. Oliveto, **Feiyang Zheng**. “Hybrid Selection Allows Steady-State Evolutionary Algorithms to Control the Selective Pressure in Multimodal Optimisation.” *Genetic and Evolutionary Computation Conference (GECCO)*, 2025. Experiments in C++, analysis in MATLAB.

SELECTED PROJECTS & AWARDS

Scientific Data Compression Without Decompression | C++, CUDA, Python | ongoing Aug 2025 – Present

- Developing lossy compression for large-scale scientific simulation data using low-rank and quantization-based methods, with sub-region access and processing directly on the compressed representation (no full decompression).
- C++/CUDA pipeline with Python tooling for parameter sweeps and evaluation on public scientific datasets.

GPU Kernel Data-Race Detector | C++, Python, CUDA | ongoing

2026 – Present

- Building a data-race detector for CUDA kernels (unpublished; details available on request).

ASC22-23 Supercomputer Challenge — First Prize | Python, TensorFlow, DeepMD, Spack

2022 – 2023

- Trained DeepMD (interatomic potential model) and ran molecular dynamics; profiled with TensorFlow Profiler.
- Configured a 4-node cluster with 8×A100-80GB-SXM4 from scratch; built the environment with Spack and Conda.

6th Asia-Pacific HPC-AI Student Competition — Second Place | Python, PyTorch, Nsight

2023

- Accelerated multi-node BLOOM LLM inference; removed inefficiencies found with Nsight and PyTorch Profiler.

TECHNICAL SKILLS

Languages: C/C++, Python, Java, MATLAB, SQL (Postgres)

Parallel & GPU: CUDA, OpenMP, SIMD/AVX, PTX/SASS, cuBLAS, LibTorch, PyTorch, TensorFlow, Horovod

Profiling & Tools: Nsight Systems/Compute, VTune, nvdisasm, cuobjdump, Git, CMake, Docker, Spack, Conda, L^AT_EX

SERVICE

Sub-reviewer: ICS 2026, SSDBM 2026, SC 2026